

Thomas Mbrice

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EDUCATION

Stony Brook University

Bachelor of Science in Computer Science

Stony Brook, New York

Expected May 2026 | GPA: 3.51/4.0

Relevant Coursework: Software Development, Advanced Algorithms, System Design, Data Structures

Honors & Scholarships: Dean's List: Fall 2022, 2023, 2024, Presidential Merit Scholarship: 2022 - 2025

EXPERIENCE

Stony Brook University, Vertically Integrated Project

Research Assistant

Stony Brook, NY

08/2024 – Present

- Engineered a large-scale autonomous vehicle simulation platform processing 100+ ethical scenarios using JavaScript and Python, which will be referenced in a future academic paper.
- Integrated real-time OpenStreetMap data across user created environments and increased speed of simulation by 200%.

Stony Brook University, College Of Engineering and Applied Science

Teaching Assistant

Stony Brook, NY

9/2024 – 12/2024

- Demonstrated interpersonal skills and articulate communication while assisting System Fundamentals to 200+ students, committed to professional development through creating comprehensive learning materials.
- Provided targeted weekly tutoring to struggling students, with 90% improving their grades by at least one letter grade.

Commonwealth Cooperative Bank

Technical Intern

Boston, MA

05/2023 – 08/2023

- Developed solution for database architecture using object-oriented design principles, showcasing innovative approaches to scalability by implementing distributed systems that reduced data retrieval time by 75%.
- Authored technical documentation and guides for 42 Python scripts, reducing employee onboarding time by at least one day and decreasing support tickets by 40%, proving problem solving ability.

PROJECTS

PhReddit, MongoDB, Express, React, Node.js, AWS, Jest

Fall, 2024

- Engineered a scalable backend using Express and MongoDB aggregation pipeline, with the ability to handle 1000+ concurrent requests with average response time under 100ms.
- Developed a Reddit Clone with React-based, user-friendly, front-end interface, including posts, communities and search.
- Secured account functionality using JWT tokens resulting in 0 security incidents.
- Containerized the application environment and test suite using Docker, enabling consistent Jest testing across development and Amazon EC2 production servers.

Lilota, MQTT, Arduino, ESP, C

Winter, 2025

- Developed and implemented an unencrypted MQTT client integration for Lilota embedded systems runtime, enabling real-time device communication and remote monitoring capabilities using Adafruit broker services.
- Contributed to open source TCL-based IoT framework by implementing networking features, allowing seamless communication between embedded devices and supporting remote data collection and device management.

HealthVizz, Flask, D3, Python, JavaScript

Fall, 2024

- Developed an interactive health analytics dashboard using D3.js and Flask, featuring 5 types of visualizations including real-time updating K-Means and Distribution selection.
- Optimized data processing by implementing Python multiprocessing, reducing calculation time for visualizations by 60%.

CodeUp, Next.js 15, Django, React, TypeScript

Summer, 2024

- Designed a production-grade code editor platform using Next.js and TypeScript, featuring real-time code execution across two coding languages (Java and Python).
- Successfully onboarded 100 users and leveraged Agile recursive software development methodologies to improve efficiency and adapt to user feedback.

TECHNICAL SKILLS

Frameworks & Tools: Node.js, Git, SQL, API/JSON, Express, MongoDB, Flask, MERN, Jest, AWS/EC2, Docker, React

Languages: Java, JavaScript/TypeScript, HTML/CSS, C/C++, Python, Bash

Certificates: Stanford Machine Learning Specialization, AWS Cloud Practitioner Essentials